

Module Core 3

Python Applications in Ocean Sciences

SAMOS2026 Core3CS Project

Visualising Remote-Sensed Ocean Colour Data

Group name

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Project Overview

This project analyses and visualises satellite-derived ocean colour data from the ESA-CCI climatology alongside bathymetry data to investigate phytoplankton distribution and seasonal variability in St Helena Bay and Cape Columbine, located within the southern Benguela upwelling system on the west coast of South Africa. By combining chlorophyll-a concentration maps, time series analysis, and high-resolution bathymetry, we aim to understand how the physical structure of the shelf drives the patterns of primary productivity observed in this highly dynamic coastal environment.

Introduction to study region

The coastal region of St Helena Bay and Cape Columbine, located on the west coast of South Africa, was selected because it forms a distinctive part of the southern Benguela upwelling system (Monteiro and Roychoudhury, 2005). Thus, it represents one of the most dynamic, biological, and ecologically important regions as it is characterised by wind-driven upwelling, where cold, nutrient-rich deep waters rise to the surface and stimulate high levels of primary productivity, particularly phytoplankton growth (Shannon, 1985). Cape Columbine, which is one of the main southern Benguela upwelling centres, often drives these upwelled waters north-eastward, onto the major embayment of the system, St Helena Bay (Pitcher et al., 2022), which serves as a retention region, where hydrographic variability strongly influences physical oceanographic conditions (such as chlorophyll concentration and the distribution of water masses along the region) (Lamont et al., 2015).

High chlorophyll concentrations and seasonal phytoplankton blooms are therefore common in this region, supporting commercially important fisheries such as sardine and anchovy (Van der Lingen et al., 2006). Microbial communities, including nano- and picoplankton also play an important role in nutrient cycling and ecosystem functioning within the bay (Probyn, 1992). Additionally, published research by Monteiro and Roychoudhury (2005) has highlighted that local bathymetry and hydrographic processes in this region also influence the distribution of sediment trace metals. The embayment bay increases the retention of sediment trace metals in its shallow areas, thus allowing the accumulation of metals in and around the bay. Moreover, recent research has underscored St Helena Bay as a key region for assessing chlorophyll concentrations using multi-satellite ocean colour analyses, because of its strong seasonal variability (Oehley et al., 2025). Altogether, these studies act as a baseline for focusing on Cape Columbine and St Helena Bay coastal region: a biogeographical region where bathymetry and hydrography influence physical and biological oceanographic processes, thus underscoring it as a good exemplary region to visualise bathymetry and ocean colour climatology.

Description of the datasets and the listings of the metadata from the bathymetry and chlorophyll data files

The bathymetry file was downloaded from the GMRT MapTool, in netCDF (CF Convention) format. The file dimension had coordinates, latitudes (min=-32.22°; max=-34.77°) and longitudes (min=16.99°; max=18.91°), and a data variable, altitude at a high grid/spatial resolution (61m/node) (Ryan et al., 2009). Additionally, the file was unmasked, synthesising multibeam sonar data with GEBCO data (Ryan et al., 2009). Thus, providing seafloor bathymetry that matched the 4 km chlorophyll climatology.

The chlorophyll-a concentration data applied in this study were obtained from the European Space Agency Climate Change Initiative (ESA-CCI) Ocean Colour monthly climatology product. This dataset consists of a merged multi-sensor satellite record with a spatial resolution of 4 km and is organized as a 12-month climatology representing long-term mean conditions for each month of the year.

The main variable, `chlor_a` (chlorophyll-a concentration), is provided in NetCDF format with units of milligrams per cubic metre (mg m^{-3}). The principal metadata dimensions include 'lon' (longitude, spanning -180° to 180°), 'lat' (latitude, spanning -90° to 90°), and time (12 monthly intervals corresponding to January–December). The dataset follows CF metadata conventions and is referenced to the WGS84 geographic coordinate system. Missing or invalid observations are identified using a standard fill value of -999.

Results

Bathymetry

The map (figure 1) demonstrates the bathymetry and topography of St Helena Bay and Cape Columbine, ranging from ~-400m-200m. Cape Columbine is a prominent landform region on the west coast while St Helena Bay, lies to the northeast, as a major embayment of the southern Benguela upwelling is in the lee of Cape Columbine (Pitcher et al., 2022). The bathymetry map underscores that shelf break, together, with coastal topography drive physical oceanographic processes, and through using map tools, such as GMRT, we can combinedly visualise how hydrography and topography drive chlorophyll biomass in and around St Helena Bay.

Bathymetry mapping of St Helena Bay and Cape Columbine coastal region

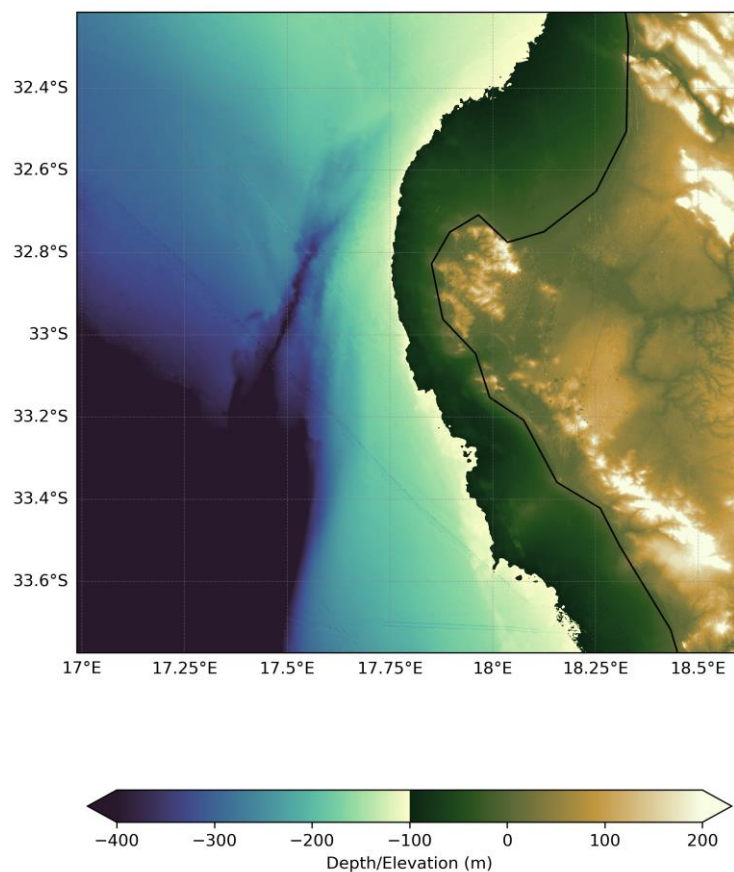


Figure 1: Bathymetry representative of St Helena Bay and Cape Columbine coastal region, distinct areas of southern Benguela upwelling system. along west coast of South Africa

Mean annual chlorophyll-a map

Figure 2 illustrates the spatial distribution of annual mean chlorophyll-a concentrations across St Helena Bay and Cape Columbine Bay, presented as a raster surface with the coastline delineated in black. A logarithmic colour scale is applied to the dataset, as it is appropriate for scaling chlorophyll-a observations. Logarithmic transformation improves the representation of both low and high concentration ranges simultaneously, preventing elevated values from disproportionately dominating the visual field and enhancing the interpretability of spatial variability (Campbell, 1995; IOCCG, 2000).

The progressive transition from dark blue to green and yellow provides a coherent representation of chlorophyll gradients while minimising the introduction of artificial visual discontinuities (Figure 2). The spatial pattern reveals a pronounced gradient in chlorophyll-a concentration from offshore to coastal areas. Lower concentrations dominate the offshore sector of the study area, represented by cooler blue tones, whereas progressively elevated concentrations occur toward the nearshore region, where green and yellow hues become increasingly prominent. Peak chlorophyll concentrations are concentrated adjacent to the coastline. This distribution is characteristic of productive eastern boundary coastal environments, where processes such as coastal upwelling, nutrient enrichment, estuarine exchange, terrestrial runoff, and enhanced vertical mixing promote elevated phytoplankton productivity in nearshore waters relative to the more oligotrophic offshore domain (Cloern and Jassby, 2010).

Mean Annual Chlorophyll-a Concentration in St Helena Bay and Cape Columbine from 1998-2020

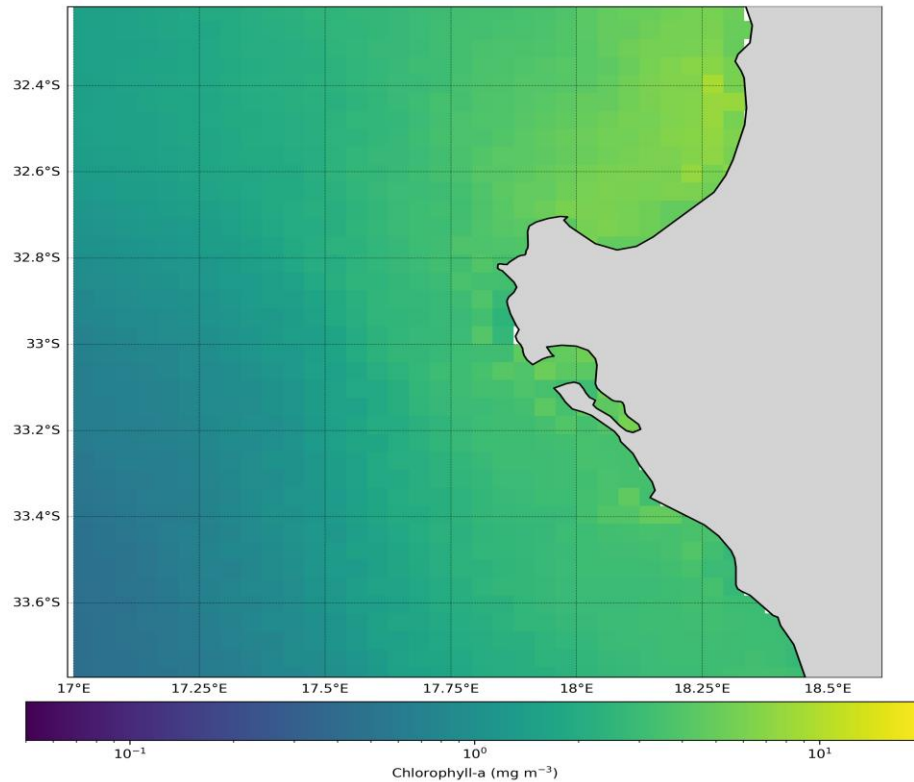


Figure 2: Map representing the mean annual chlorophyll-a concentration in St Helena Bay and Cape Columbine from 1998-2020

Monthly Chlorophyll-a Climatology

Looking at Figure 3, we can see from the twelve monthly panels that chlorophyll-a concentration across St Helena Bay and Cape Columbine shifts quite noticeably throughout the year, with the coastal waters reaching their highest values during March and April when the Benguela upwelling system is at its most active and nutrient-rich water is being brought to the surface consistently, while by July and August those same coastal areas quiet down to their lowest values as the winds ease off and thermal stratification cuts off the nutrient supply from below. What stands out particularly is that the nearshore strip closest to Cape Columbine stays elevated for longer into winter than the rest of the region, which points to the narrow shelf there continuing to drive localised upwelling even after the broader regional signal has already weakened.

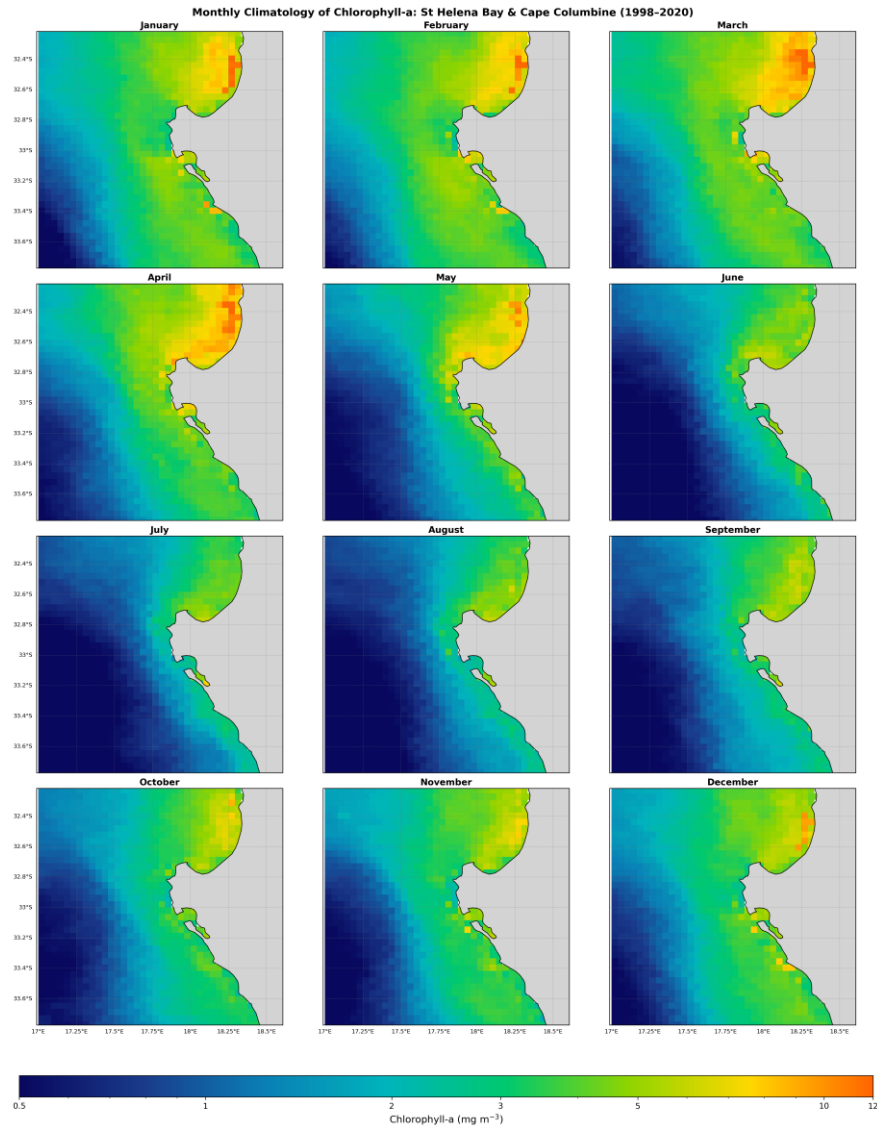


Figure 3: Monthly climatology of chlorophyll-a concentration (mg m^{-3}) across St Helena Bay and Cape Columbine, South Africa.

Seasonal Time Series

From Figure 4 we can see that the regional mean peaks at just over 5 mg m^{-3} in March before dropping to around 2.2 mg m^{-3} by July, which lines up well with the seasonal story told by the monthly maps. The single grid point tracks that same rise and fall but sits consistently below the regional mean by roughly 1 to 2 mg m^{-3} for most of the year, pointing to the most productive waters being concentrated in a narrow coastal band rather than spread evenly across the whole region. Perhaps most telling is the way the two lines draw closer together through summer, suggesting that once

productivity collapses the spatial differences within the region largely disappear and the whole area becomes uniformly low.

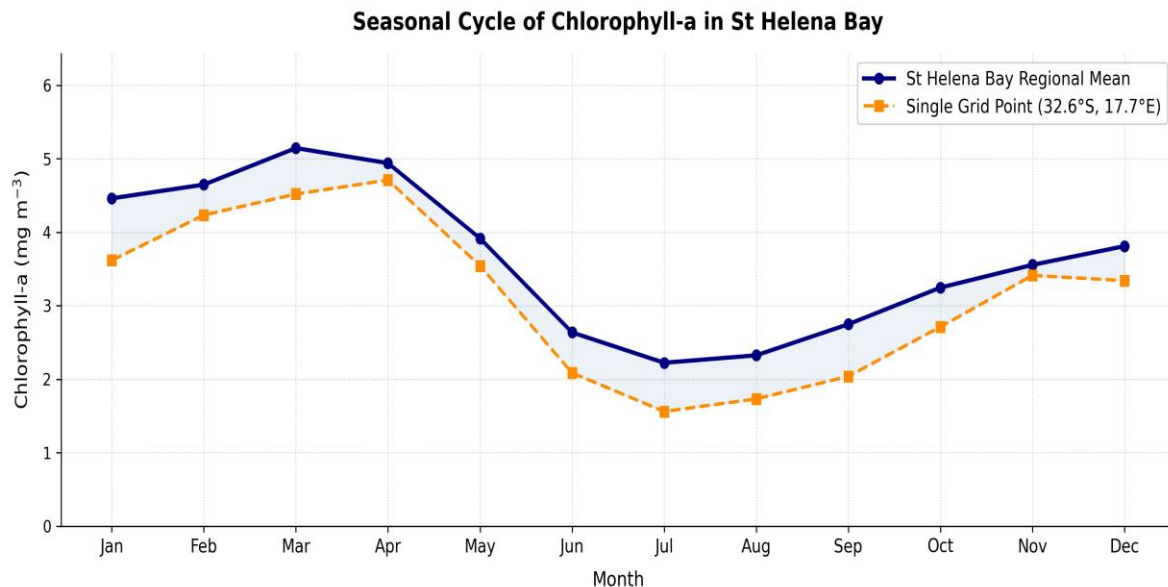


Figure 4: Seasonal cycle of chlorophyll-a concentration (mg m^{-3}) in St Helena Bay (17.5° – 19.0°E , 33.5° – 32.0°S) derived from ESA-CCI Ocean Colour climatology (1998–2020).

AI Disclaimer

Microsoft Copilot AI was used to assist with refining the descriptions of the plots/maps generated in the project, as well as resolving Git errors encountered during submission.

- Monthly chlorophyll-a maps and time series: Example prompts used include "how do I fix: error: You have not concluded your merge". All data processing, coding, and scientific interpretation were carried out independently, with AI supporting the writing and technical workflow only. Any AI-generated text was reviewed and adjusted to accurately reflect the results before being included in this work.
- Bathymetry mapping: Example prompt "I am currently generating a large figure with the map of the region bathymetry, using an adequate colormap. I previously used terrain to represent my colour, but it does not represent bathymetry and topography well. Which colour map would be scientifically correct in the

oceanography field to use as bathymetry”. This is **not ideal for bathymetry**, because “turbo” is a perceptually uniform colormap, but oceanography usually uses **scientific colormaps** (e.g., `cmocean.cm.bathy` or `viridis`) to represent depth.

- Example prompt: “I am getting an error when I use `cmocean` package, `ModuleNotFoundError: No module named 'cmocean'`”. The error means you don’t have the `cmocean` package installed. It’s a specialized colormap library for oceanography, but it’s optional — you can either install it or use built-in Matplotlib colormaps instead.
- Mean annual chlorophyll-a mapping: Example prompt: “I am getting a `'ValueError: Cannot add feature because projection is not defined'` error with `cartopy`. Can you help me understand what I'm missing?” The projection was not defined when creating the matplotlib axes. The `GeoAxes` object did not know how to transform the geographic coordinates onto the plot, making it impossible to add features like land masses, coastlines, or borders.

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